Drawing and Using Informal Lines of Fit
Answer Key
Grade 8 / Pre-Algebra

Quick Answers

1. the greatest plotted value = 28, —
2. 3
3. the slope of the line = 3, the predicted value at 7 = 26
4. the mean of the plotted values = 14, —
5. -3
6. the slope of the line = 4, the predicted value at 10 = 44
7. the slope = 3, the value at 8 = 31, —
8. 90
9. the predicted score at 4 hours = 72, —
10. the slope = -3, the predicted value at 9 = 5, —
11. the prediction at 6 = 27, —
12. the predicted pulse at 25 minutes = 68, the change over 10 extra minutes = -4, —

What is verified

5 of 12 problems fully machine-checked against SymPy.
— marks an answer only you can judge — problems 1, 4, 7, 9, 10, 11, 12. The worked solution states what a correct response must contain.

Curriculum

Level: Grade 8 / Pre-Algebra
8.SP.A.1 – *problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12*
Difficulty 1–5 of 5 across 24 tagged checks.

1. Fundraiser plot: give the greatest number of donations, then draw a line of fit.

(a) Solution: The topmost point is the week with 7 posters. $\boxed{28}$

(b) Line of fit (instructor-judged). A correct line is one straight ruler line running from lower left to upper right through the middle of the cloud, with about as many points above it as below and none far away from it. A line joining only the first and last points, a curve, or a line drawn along one edge of the cloud does not earn full credit.

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2. A line of fit passes through $(1, 6)$ and $(5, 18)$. Find its slope.

Solution: Take the two points that lie exactly on the drawn line, not two data points: $m = \dfrac{18-6}{5-1} = \dfrac{12}{4}$. $\boxed{m = 3}$

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3. A line of fit passes through $(0, 5)$ and $(4, 17)$. Find its slope and predict at $x = 7$.

Solution: $m = \dfrac{17-5}{4-0} = \dfrac{12}{4} = 3$. The line meets the vertical axis at 5, so its equation is $y = 3x + 5$, and at $x = 7$ that gives $3(7) + 5 = 21 + 5$.

$$\boxed{m = 3 \qquad y = 26 \text{ at } x = 7}$$

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4. Savings plot: find the mean of the plotted values, then draw a line of fit through the middle of the data.

(a) Solution: Add the six values and divide by six:

$$\frac{5 + 8 + 12 + 15 + 19 + 25}{6} = \frac{84}{6}$$

$$\boxed{14}$$

(b) Line of fit (instructor-judged). A good line rises from left to right and passes close to the middle of the data — around the middle month, at a height near the mean of 14 — with a similar number of points on each side. A line drawn only through the first and last points tilts too steeply and does not earn full credit.

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5. A line of fit passes through $(2, 30)$ and $(8, 12)$. Find its slope.

Solution: $m = \dfrac{12-30}{8-2} = \dfrac{-18}{6}$. The negative sign matches the picture: the line falls as you move right.

$$\boxed{m = -3}$$

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6. A line of fit passes through $(1, 8)$ and $(6, 28)$ and crosses the vertical axis at 4. Find the slope and predict at $x = 10$.

Solution: $m = \dfrac{28-8}{6-1} = \dfrac{20}{5} = 4$, so the equation is $y = 4x + 4$. At $x = 10$: $4(10) + 4 = 40 + 4$.

$$\boxed{m = 4 \qquad y = 44 \text{ at } x = 10}$$

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7. A line of fit passes through $(0, 7)$ and $(5, 22)$. Find the slope, write the equation, and evaluate at $x = 8$.

Solution: $m = \dfrac{22-7}{5-0} = \dfrac{15}{5} = 3$. The first point is on the vertical axis, so $b = 7$ can be read straight off. At $x = 8$: $3(8) + 7 = 24 + 7$.

$$\boxed{m = 3 \qquad y = 31 \text{ at } x = 8}$$

(b) Equation (instructor-judged). The equation is $y = 3x + 7$. Accept an equivalent rearrangement. An equation with the 3 and the 7 swapped, or with the intercept missing, is wrong even if the slope was found correctly.

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8. $y = 6x + 48$. Predict the score after 7 hours of revision.

Solution: Substitute the hours for x: $6(7) + 48 = 42 + 48$. $\boxed{90}$

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9. $y = 6x + 48$. Predict at 4 hours, and say what the slope means.

(a) Solution: $6(4) + 48 = 24 + 48$. $\boxed{72}$

(b) Interpretation (instructor-judged). The slope 6 is a rate: according to the line, each extra hour of revision goes with about 6 more marks on the test. Accept any wording that attaches the 6 to one extra hour and to marks. Calling it only "the steepness" misses the context and earns partial credit.

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10. A line of fit passes through $(2, 26)$ and $(6, 14)$. Find the slope, the equation, and the value at $x = 9$.

Solution: $m = \dfrac{14 - 26}{6 - 2} = \dfrac{-12}{4} = -3$. Substituting the point $(2, 26)$ gives $26 = -3(2) + b$, so $b = 32$. At $x = 9$: $-3(9) + 32 = -27 + 32$.

$$\boxed{m = -3 \qquad y = 5 \text{ at } x = 9}$$

(b) Equation (instructor-judged). The equation is $y = -3x+32$; accept an equivalent rearrangement. A positive slope, or an intercept that does not fit both given points, is wrong.

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11. Bean plant, days 1 to 10, line of fit $y = 2.5x + 12$. Predict day 6, then judge a prediction for day 40.

(a) Solution: $2.5(6) + 12 = 15 + 12$. $\boxed{27}$

(b) Caution (instructor-judged). The measurements only cover days 1 to 10, so day 40 is far outside the range the line was fitted to. Nothing in the data says the plant keeps growing at the same steady rate for another month — real plants level off. Accept any wording that contrasts day 40 with the range of the collected data.

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12. Exercise against resting pulse, line of fit $y = -0.4x + 78$. Predict at 25 minutes, give the change per 10 minutes, and judge causation.

Solution: At 25 minutes: $-0.4(25) + 78 = -10 + 78 = 68$. The slope is the change per single minute, so over 10 extra minutes the change is -0.4×10.

$$\boxed{\text{pulse} = 68 \qquad \text{change} = -4}$$

(c) Judgement (instructor-judged). A correct response says the line shows only an association between exercise time and resting pulse in this class, and that an association is not proof of cause. Students who are already fitter may both exercise more and have lower pulses, and no experiment was carried out. Accept any wording that separates association from cause.